

EECE 571Q

Lecture 05: Classical codes, CSS codes, and bounds on QECC

Tuesday 26 May 2026

Announcements

- Quiz 2 today
- More supplementary exercises posted
- Project information on PrairieLearn - choose partners by 2 June, topic proposal due 12 June

Last time

We defined $[[n, k, d]]$ stabilizer codes.
distance

The 9-qubit Shor code is a $[[9, 1, 3]]$ code.
log. physical

Name	Operator
g_1	$ZZIIIIIIII$
g_2	$IIZZIIIIII$
g_3	$IIIZZIIII$
g_4	$IIIIIZZIII$
g_5	$IIIIIIZZI$
g_6	$IIIIIIIZZ$
g_7	$XXXXXXIII$
g_8	$IIIXXXXXX$
$\tilde{Z}$	$XXXXXXXXXX$
$\tilde{X}$	$ZZZZZZZZZZ$

n phys.

k logical qubits

stabilizer gens:
 $n - k$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Last time

We defined code distance and conditions for errors to be correctable or undetectable w.r.t. stabilizer generators.

$$\hat{N}(S) = \{N \mid [M, N] = 0 \ \forall M \in S\}$$

Gottesman Thm 3.4

“The set of undetectable errors for a stabilizer code S is $\hat{N}(S) \setminus \hat{S}$. The **distance** of S is $\min\{\text{wt}(E) \mid \hat{E} \in \hat{N}(S) \setminus \hat{S}\}$ ”.

Gottesman Thm 3.5

“The stabilizer code S corrects a set of errors $\mathcal{E} \subseteq \mathcal{P}_n$ iff $\hat{E}^\dagger \hat{F} \notin \hat{N}(S) \setminus \hat{S}$ for all $E, F \in \mathcal{E}$ ”.

Last time

We defined classical error correcting codes.

Source: Gottesman, Def. 4.1

"A classical error-correcting code $(e, \mathcal{E})$ is a **map** $e : [1 \dots K] \rightarrow [1 \dots M]$ with the set of **correctable errors** $\mathcal{E}$ (consisting of functions $E : [1 \dots M] \rightarrow [1 \dots M]$) with the following property: $\exists$ function $d : [1 \dots M] \rightarrow [1 \dots K]$ such that $\forall E \in \mathcal{E}, \forall x \in [1 \dots K],$ "

$$d(E(e(x))) = x$$

Learning outcomes

Today:

- 1 Define classical linear codes, and formulate them in terms of generator and parity check matrices
- 2 Express Pauli group elements using binary symplectic representation, and describe the relationship between stabilizers and linear codes
- 3 Use classical linear codes to construct CSS codes (Shor code and Steane code)
- 4 Determine the smallest possible code that can correct arbitrary single-qubit errors.

Classical codes

The *distance* of a code $C \subseteq \mathbb{Z}_N$ is

$$\min \{ \text{wt}(E) \mid \exists x \neq y \in C, Ex = y \}$$

Consequence: a distance d code

- corrects general errors on up to $\lfloor (d-1)/2 \rfloor$ coordinates
- corrects erasure errors on up to $d-1$ coordinates
- detects general errors on up to $d-1$ coordinates

Notation: (n, K, d) code

$$[0 \ 0 \ - \ 0 \ 0]$$

Classical linear codes

Source: Gottesman, Def. 4.3, section 4.2

A code $C \subseteq \mathbb{Z}_2^n$ is a *linear code* if

$$x, y \in C \rightarrow x+y \in C$$

Notation: $[n, k, d]$ code. $[[n, k, d]]$
stabilizer codes

Codewords are linear combinations of basis states for (sub)space C .

Example: repetition code. $k = 1$ basis state, $2^k = 2$ codewords.

$$x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad x+x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \pmod{2}$$

Classical linear codes

Given k basis vectors $x_1, \dots, x_k$, define a $k \times n$ **generator matrix**

$$G_C = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} \begin{matrix} \updownarrow k \\ \leftarrow n \rightarrow \end{matrix}$$

The linear map $v \in \mathbb{Z}_2^k \rightarrow G_C^T v \in \mathbb{Z}_2^n$ is the *encoder* for C .

Example: repetition code.

$$x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$G_C = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$$

$$G_C^T (0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} (0) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$G_C^T (1) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} (1) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Classical linear codes

Given G_C , a $(n - k) \times n$ **parity check matrix** H_C contains rows y_i such that $Gy_i = 0$, with $\{y_i\}$ a maximal linearly independent set.

Relationship between G_C and H_C : $G_C H_C^T = H_C^T G_C = 0$

Example: repetition code.

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$G H^T = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \pmod{2}$$

Classical linear codes

How to model errors?

$$x \rightarrow x + e$$

Apply the parity check matrix...

$$H_c (x + e) = H_c \underbrace{x}_{G^T v} + H_c e = \underbrace{H_c G^T}_{} v + H_c e = H_c e$$

Example: repetition code.

$$x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad e = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$H_c e = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ syndrome}$$

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

The Hamming code

Suppose we wish to correct a 1-bit error on 2^{r-1} bit code words. Need all error syndromes to be unique.

$$H_c = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$r=3$

2^{r-1} columns

$$H_c \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$H_c \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$[n, k, d]$

Code parameters: $[7, 4, 3]$

General parameters: $[2^{r-1}, 2^{r-1}-r, 3]$

$$G = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Dual codes

Given code C , its dual code $C^\perp$ is

$$C^\perp = \{ y \in \mathbb{Z}_2^n \mid x \cdot y = 0 \quad \forall x \in C \}$$

The dual code exchanges roles of G and H .

If C has code parameters $[n, k, d]$, $C^\perp$ has parameters $[n, n - k, d]$.

$$G_C = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$H_C = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$[7, 3, 4]$$

Relationship with stabilizer codes

Binary symplectic representation: isomorphism between $\hat{\mathcal{P}}_n$ and vectors in $\mathbb{Z}_2^n \times \mathbb{Z}_2^n$.

$$\rho_1: \quad I = (0 \mid 0) \quad X = (1 \mid 0) \quad Z = (0 \mid 1) \quad Y = (1 \mid 1)$$

$$X^a Z^b = (1 \mid 0 \mid 0 \mid 1 \mid 1) \quad X^{a_0} Z^{b_0} \otimes X^{a_1} Z^{b_1} \otimes \dots = (a_0 \ a_1 \dots \mid b_0 \ b_1 \dots)$$

Commutator computed from symplectic inner product:

$$P_1 = (\vec{X}_1 \mid \vec{Z}_1) \quad P_2 = (\vec{X}_2 \mid \vec{Z}_2)$$

$$\vec{V}_1 \odot \vec{V}_2 = \vec{X}_1 \cdot \vec{Z}_2 + \vec{Z}_1 \cdot \vec{X}_2 \pmod{2}$$

We can represent stabilizers in this format:

$$\begin{aligned} ZZI &= (0 \ 0 \ 0 \mid 1 \ 1 \ 0) \\ IZZ &= (0 \ 0 \ 0 \mid 0 \ 1 \ 1) \end{aligned}$$

H: parity check matrix of 3-bit repetition code!

Relationship with stabilizer codes

(Source: Gottesman Thm. 4.4)

More generally, given a linear code C and parity check matrix H , define stabilizer code with binary symplectic representation,

$$(0 \mid H)$$

This stabilizer code has the same basis codewords as C (plus linear combinations thereof), and corrects the same set of errors.

(Source: Gottesman Def. 5.1)

↗ *Calderbank, Shor, Steane*

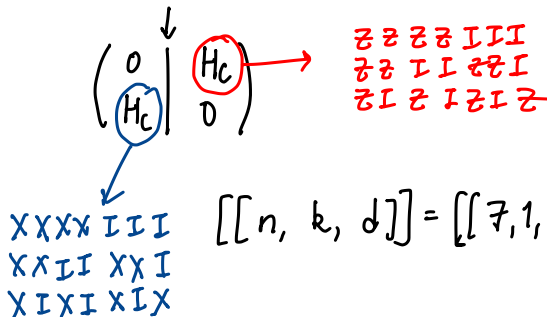
A stabilizer code is a CSS code if there exists a choice of stabilizer generators with binary symplectic representation of the form

$$\begin{pmatrix} 0 & | & A \\ B & | & 0 \end{pmatrix}$$

The Steane code

Example: choose both A and B to be H of $[7, 4, 3]$ Hamming code:

$$H_c = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$



homework

What are the logical operations? What are the code parameters?

The Steane code: $[[7, 1, 3]]$

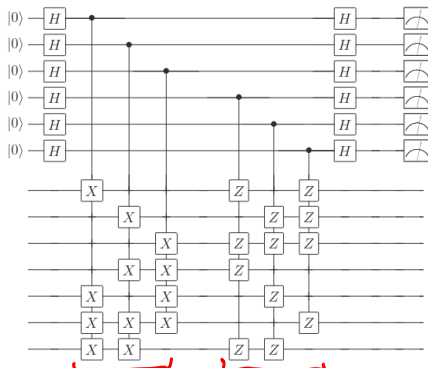


Figure 10.16. Quantum circuit for measuring the generators of the Steane code, to give the error syndrome. The top six qubits are the ancilla used for the measurement, and the bottom seven are the code qubits.

Screenshot: Nielsen & Chuang, chapter 10.5.8.

CSS codes, more formally

(Source: Gottesman Thm. 5.1)

Let C_1 be an $[n, k_1, d_1]$ classical linear code with parity check matrix H_1 , and C_2 be an $[n, k_2, d_2]$ classical linear code with parity check matrix H_2 , with $C_1^\perp \subseteq C_2$.

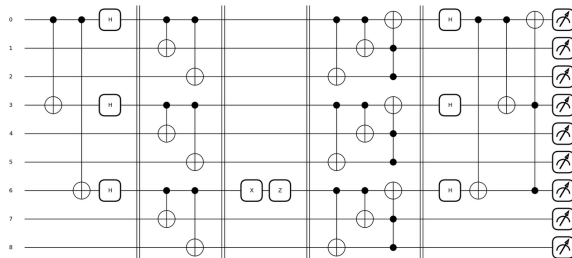
The stabilizer code (CSS) code with stabilizer generators

$$\left(\begin{array}{c|c} 0 & H_1 \\ H_2 & 0 \end{array} \right)$$

is an $[[n, k, d]]$ quantum code with $k = k_1 + k_2 - n$ and $d \geq \min(d_1, d_2)$.

Shor code

The Shor code is also a CSS code.



Name	Operator
g_1	$ZZIIIIIIII$
g_2	$IZZIIIIIIII$
g_3	$IIIZZIIIIII$
g_4	$IIIIIZZIIII$
g_5	$IIIIIIIZZII$
g_6	$IIIIIIIIIZZ$
g_7	$XXXXXXIIII$
g_8	$IIIXXXXXXX$
Z	$XXXXXXXXXX$
X	$ZZZZZZZZZZ$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

The Hamming bound

What is the smallest (non-degenerate) code that corrects arbitrary single-qubit errors?

General bound:

- n physical qubits
- k logical qubits
- up to t errors

★ We will start here next time.

If a non-degenerate code exists,

The smallest code: $[[5, 1, 3]]$

$$|0_L\rangle = \frac{1}{4} \left[|00000\rangle + |10010\rangle + |01001\rangle + |10100\rangle \right. \\ \left. + |01010\rangle - |11011\rangle - |00110\rangle - |11000\rangle \right. \\ \left. - |11101\rangle - |00011\rangle - |11110\rangle - |01111\rangle \right. \\ \left. - |10001\rangle - |01100\rangle - |10111\rangle + |00101\rangle \right]$$

$$|1_L\rangle = \frac{1}{4} \left[|11111\rangle + |01101\rangle + |10110\rangle + |01011\rangle \right. \\ \left. + |10101\rangle - |00100\rangle - |11001\rangle - |00111\rangle \right. \\ \left. - |00010\rangle - |11100\rangle - |00001\rangle - |10000\rangle \right. \\ \left. - |01110\rangle - |10011\rangle - |01000\rangle + |11010\rangle \right]$$

Name	Operator
g_1	$XZZXI$
g_2	$IXZZX$
g_3	$XIXZZ$
g_4	$ZZIXZ$
$\bar{Z}$	$ZZZZZ$
$\bar{X}$	$XXXXX$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

The quantum Gilbert-Varshamov bound

Quantum Hamming bound tells us parameters of smallest code.
What about existence criteria for *any* code?

(Source: Gottesman Thm. 7.1)

There exists an $((n, 2^k, d))$ QECC if

Bound also proves existence of an $[[n, k, d]]$ stabilizer code.

The quantum singleton bound (Knill-Laflamme bound)

(Source: Gottesman Thm. 7.3)

If there exists an $((n, q^k, d))_q$ QECC exists (qudit dim. q),

Gottesman, p. 112: quantifies “how well information can be spread around without accidentally repeating anything”.

See textbook for proof.

Next time

Next class (last class):

- The Clifford group
- Universal gate sets and logical operations
- Efficient classical computation of Clifford circuits

Action items:

- ① Select project partner and think about topics
- ② Supplementary exercises

Recommended reading:

- From this class: Gottesman Chs. 4, 5, 7; N&C 10.1-10.5
- For next class: Gottesman Ch. 6, N&C 4.5, 10.6